

Research papers

Model-data fusion domain adaptation for battery State of Health estimation with fewer data and simplified feature extractor

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ABSTRACT

Accurate estimation of State of Health (SOH) with fewer data under different operation conditions is challenging for the lithium-ion battery. In this paper, a novel model-data fusion domain adaptation is initiated for SOH estimation. Distinguishing from existing model-based methods and data-driven methods, the proposed method not only can improve the accuracy of estimation with reliability and interpretability, but also can adapt to the different operation conditions with fewer data. Specifically, through introducing the battery models, more reliable and interpretable features are generated according to operation conditions and the layers of the feature extractor can be reduced. With the model-based features, the data-driven margin disparity discrepancy (MDD) based domain adaptation is introduced to transfer knowledge between the dataset of different operation conditions by reducing the probability distribution discrepancy. Therefore, the estimation performance with fewer training data in the target domain is improved with reliable and interpretable features, and with the transfer of knowledge learned from the source domain under various operation conditions. The effectiveness of the proposed method is verified by the NASA dataset and our experiment on the FST battery. The results demonstrate the estimation performance can be improved with fewer data and the reduction of layers in the feature extractor.

1. Introduction

Lithium-ion batteries are one of the most important components of the energy storage system [1]. Owing to the high energy density, long cycle life and low self-discharge rate, the lithium-ion battery has been applied in electricity systems such as electric vehicles, smart grids and cell phones [2]. Nevertheless, the capacity and internal resistance of the battery will change over time of usage which threatens the performance of the battery [3,4]. Hence, monitoring the battery state of health (SOH) under the battery management system (BMS) is important for improving overall performance, extending battery life and preventing self-ignition [5].

In recent years, many strategies have been studied to measure the battery SOH. The estimation methods are mainly divided into two categories which are model-based methods and data-driven methods [6]. Model-based methods have been developed based on battery models such as mechanism models, equivalent models, and empirical models [7]. The development of the battery model is dependent on expert knowledge and high-accuracy parameter identification which is difficult to be derived under different operation conditions. To avoid the

drawbacks of model-based methods, data-driven methods are developed which became one of the research focuses. Without the knowledge of the complex battery reaction, the SOH could be estimated based on a large number of data. The data-driven methods include support vector machine [8], Artificial Neural Network [9], Gaussian process regression [10] and other machine learning methods [11]. Data-driven methods have been widely applied in battery state estimation and more accurate results are obtained in recent years.

Nevertheless, some challenges remain open for traditional data-driven SOH estimation methods. Firstly, the high-accurate results can be obtained based on sufficient amount of data. In the real application, the accurate and reliable label of SOH for a battery is not always available considering the complex operation conditions and specific instrument requirements [12]. Therefore, obtaining accurate estimation results with fewer training samples is critical for the real battery system. Secondly, implementing direct-monitored health features such as voltage, current and temperature as the input could obtain high accuracy in some specific conditions [13]. Nevertheless, the estimation results would be inaccurate by using untreated direct-monitored health features when the operation conditions are different [14]. Improving the

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adaptability of estimation under various operation conditions is necessary for SOH estimation. Finally, deep learning methods are developed for the SOH estimation that would be able to contribute to obtaining accurate and reliable results [11,15]. However, the ideal results of deep learning rely on complex networks and multiple layers which may also result in overfitting, high computation costs and great memory space. For the real application, designing a simplified data-driven method for obtaining estimation accuracy has practical significance at present.

To improve the estimation performance with fewer data, the transfer learning of fine-tuning is designed to estimate battery states with fewer data [16]. However, the conditions considered in [16] are limited. A series of transfer learning methods including domain adaptation, domain-adversarial neural network and multisource domain adaption are utilized in SOH estimation [17,18]. These studies show better estimation results with a few data under different operation conditions. Nevertheless, more layers and complex networks are required for extracting the SOH degradation information. Therefore, designing a simplified transfer learning method to adapt to complex operation conditions with fewer data is necessary for SOH estimation.

In this paper, a model-data fusion domain adaptation method has been proposed. The proposed method not only can improve the accuracy of estimation with reliability and interpretability, but also can adapt to the various operation conditions with fewer data. Specifically, the accuracy of estimation is sensitive to noise for fewer training data [19]. Besides, the interpretable and reliable health features are critical to improving the estimation performance [17,20]. Inspired by [21–23], different battery models are fused into the domain adaptation to obtain the noise-reduced, interpretable and reliable features. By introducing battery models, model-based features could be generated to improve the estimation performance. Besides, the layers of the feature extractor could be reduced because of the interpretable and reliable features. The detailed reasons are presented below:

1. Battery models such as the OCV calibration model could be introduced to provide features that could filter the unfavorable noise based on interpretable and expert model-based knowledge.
2. Battery features could be constructed by model-based knowledge. As a result, the effective construction of sufficient and interpretable health features from battery models under different operation conditions could provide effective and explainable SOH degradation information to improve estimation performance.
3. Various battery models provide more relevant and reliable features which are effective for reflecting the battery SOH degradation. The deep layers and complex networks of the feature extractor are not required for extracting battery SOH degradation. Hence, the exact results could be obtained with the simplified neural networks of the layers reduced feature extractor.

With the noise-reduced, reliable and interpretable features from battery models, the margin disparity discrepancy (MDD) based domain adaptation is implemented to reduce the probability distribution discrepancy between the dataset of different operation conditions. As a result, the estimation performance could be improved with the transfer of knowledge learned from the source domain where the operation conditions are different. Thus, the desirable estimation results with a few data under various operation conditions could be obtained by the proposed method. The main contributions of this paper are summarized as follows.

1. A novel model-data fusion domain adaptation for estimating battery SOH with fewer data under different operation conditions is initiated.
2. Noise-reduced features are generated by the OCV calibration model for training our method.
3. More reliable and higher interpretable features are provided based on model-based knowledge.

4. The layers of the feature extractor are reduced with the reliable features extracted by battery models.

The remainder of this article is organized as follows. Section 2 presents the introduction of health features and battery models. In Section 3, the model-data fusion domain adaptation method is introduced. The effectiveness of model-data fusion domain adaptation is verified in Section 4. In Section 5, the analysis and discussion of the results are given. Finally, the conclusion is provided in Section 6.

2. Introduction of health features and battery models

2.1. Definition of SOH and introduction of health features

The failure of a battery can be evaluated by the SOH [24]. From [25], SOH is defined by the percentage of the nominal capacity and the calculation form is commonly formulated as below:

$$SOH = \frac{C_{\text{Remaining}}}{C_{\text{New}}} \times 100\% \quad (1)$$

where $C_{\text{Remaining}}$ is the maximum capacity at present and C_{New} is the nominal capacity of the battery.

During the charging and discharging phase, some health features could be direct-measured which are the basic health features for training. The basic health features are analyzed as follows:

Current is represented by I which could influence battery life. Different current rates would influence the battery polarization. In general, the over-high current rate would result in rapid lithiation/delithiation, fast swelling, pulverization and disconnect from the current collector which would influence the homogeneity of the cathode electrolyte interface. Finally, the thickening of the solid electrolyte interphase of the anode material and the degradation of the cathode material would result in high internal resistance and accelerate the battery failure at the over-high current rate. As a result, the battery current rate cannot be over-high to protect the battery.

Voltage is represented by U which is the crucial indicator for battery degradation. For the lithium-ion battery, the rated voltage, terminal voltage and end voltage are the most common indicators which would be varied for different types of batteries. For battery safety, the lithium-ion battery operates in a certain range to avoid the erosion of metal plates and the explosion of the battery.

Temperature is represented by T . The lithium-ion battery is undesirable to operate in over high temperature because of electrolyte volatilization which would influence the ionic conductivity and viscosity of the electrolyte or in over low temperature which would reduce battery life. For the NASA dataset, batteries #29-32 and #41-44 are not operated under room temperature. From the experiment results of the NASA dataset, the battery life is decreased under high temperature and low temperature.

Time is represented by t which is recorded by the BMS. During the charging process and discharging process, a battery could charge and discharge for a long period of time.

Capacity regeneration is the phenomenon where the battery SOH could be regenerated with a long rest time which is the difficulty for battery state estimation.

2.2. Introduction of battery models

(I) Mechanism model:

Mechanism models are built to describe the system based on physical principles [26]. The battery is a chemical energy storage device and the work model is developed to reflect the ability of energy conversion [27]. During the discharging process, the chemical energy would convert to electrical energy. During the charging process, the electrical energy

would convert to chemical energy [28]. The energy conversion could be described by the work and the ability of work is varied with the battery degradation. A form of work model could be given by:

$$W = \int_{t_0}^{t_f} U(t)I(t)dt \quad (2)$$

where W is the work, $U(t)$ is the voltage at time t , $I(t)$ is the current at time t , t_0 and t_f are the initial time and ending time.

As the SOH degradation, the work in the same period is increased during the constant current mode of charging process.

(II) Incremental capacity analysis based model

Incremental capacity analysis (ICA) and differential voltage analysis (DVA) techniques have been studied to investigate battery degradation mechanisms. Through transforming the cell current and voltage into the incremental capacity (IC) and differential voltage (DV) curves, the process of lithiation and delithiation for active materials could be described [29]. The modeling of the IC curve and DV curve could be described as follows:

$$IC = dQ/dU \approx \Delta Q/\Delta U \quad (3)$$

$$DU = dU/dQ \approx \Delta U/\Delta Q \quad (4)$$

where IC denotes the increment capacity, DU denotes the differential voltage, Q denotes the battery capacity, U denotes the voltage.

The IC curve is developed to measure the effect of material activation and polarization. From the ICA of the NASA dataset under room temperature, the distribution around different peaks could be simply regarded as Gaussian distribution. As the SOH degradation, mean and variance from Gaussian distribution are increased.

(III) Equivalent model:

Although mechanism models are applied in the battery system with interpretable advantages, the requirement of initial value and sufficient knowledge are drawbacks for practical applications. Hence, equivalent circuit models (ECM) are developed for the battery state evaluation [30]. The representative models are the Thevenin model and the n-RC models [31]. In this paper, the dynamic behavior of the lithium-ion battery is illustrated by the Thevenin equivalent model which is shown in Fig. 1.

From [32], the voltage for RC circuit $U_c(t)$ is given by:

$$U_c(t) = U_c(0)e^{-\frac{Q(t)}{I(t)R_1C_1}} + R_1I(t)\left(1 - e^{-\frac{Q(t)}{I(t)R_1C_1}}\right) \quad (5)$$

where R_1 and C_1 denote resistance and capacitance for RC circuit, $I(t)$ denotes the current at time t , $Q(t)$ denotes the capacity at time t , $U_c(t)$ denotes the voltage for RC circuit at time t . At any arbitrary time, the

open circuit voltage could be given based on Kirchhoff circuit laws as:

$$OCV = V + (R_1I - U_{c0})\left(e^{-\frac{Q}{R_1C_1}}\right) - I(R_0 + R_1) \quad (6)$$

where OCV denotes open circuit voltage, V denotes terminal voltage, U_{c0} denotes initial polarization voltage.

(IV) OCV calibration model:

Equivalent models and mechanism models could describe the relationship between the OCV and capacity. However, the battery state variation is a complex phenomenon including phase transition and staging phenomena which cannot be well described by simply applying a battery model [30]. Besides, the requirement of the initial value is also a problem for an equivalent model. Hence, the lumped thermal model based Arrhenius form is considered for OCV model calibration. Arrhenius form could be given based on the temperature-based electrochemical reaction as [33]:

$$OCV = e^{\frac{E_a}{K}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)} \sum_{k=0}^m a_k (SOC)^k \quad (7)$$

where K denotes the gas constant, E_a denotes the activation energy, T_{ref} denotes the reference temperature, T denotes the temperature at present, m denotes model order, a_k denotes the reference parameter, SOC denotes the state of charge (SOC), OCV denotes open circuit voltage.

From [34], the battery capacity is related to SOC and SOC is defined as:

$$SOC = SOC_0 + \frac{Q}{Q_{max}} \quad (8)$$

where Q denotes the capacity at present, Q_{max} denotes the maximum capacity and SOC_0 denotes the initial SOC.

Through combining Eqs. (5)–(8), the terminal voltage is given by combining the form of the equivalent model and Arrhenius form which is shown below:

$$\begin{aligned} V &= e^{\frac{E_a}{K}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)} \sum_{k=0}^m a_k \left(SOC_0 + \frac{Q}{Q_{max}}\right)^k - (R_1I - U_{c0})\left(e^{-\frac{Q}{R_1C_1}}\right) + I(R_0 + R_1) \\ &= \sum_{k=1}^m w_k Q^{p_k} + p_{m+1}e^{p_{m+2}Q} + p_{m+3} \end{aligned} \quad (9)$$

where p_k and w_k denote parameters which can be considered as constants in the same cycle. In each cycle, the parameters could be identified by developing a nonlinear optimization problem which is solved by the Gauss-Newton method in this paper.

With the battery health degradation, the voltage at the same capacity

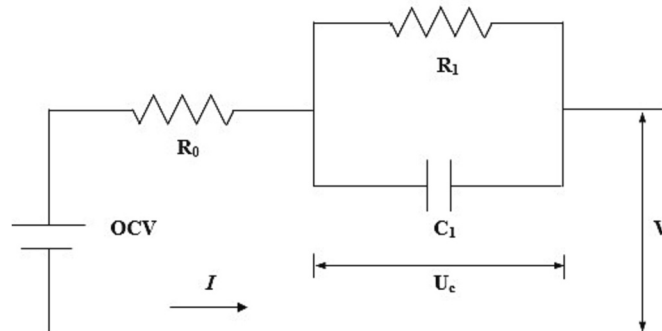


Fig. 1. Thevenin equivalent model.

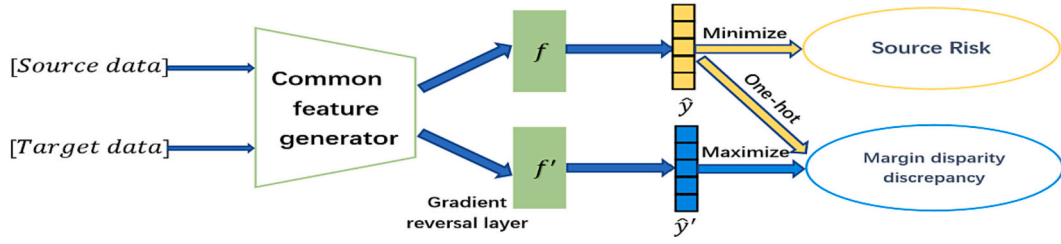


Fig. 2. Framework for MDD-based domain adaptation.

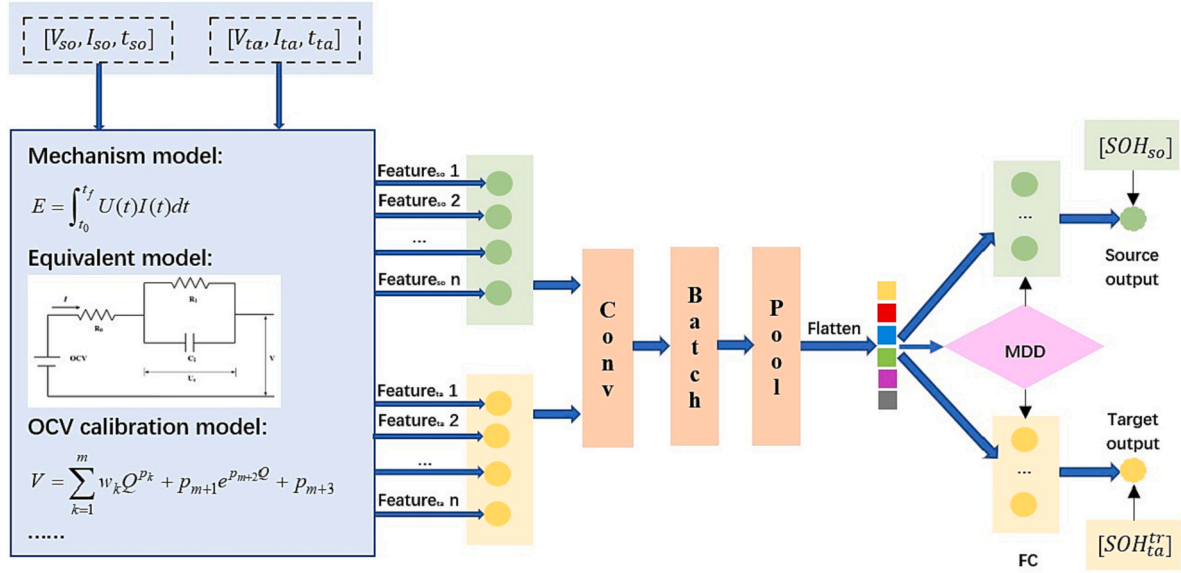


Fig. 3. Framework of model-data fusion domain adaptation.

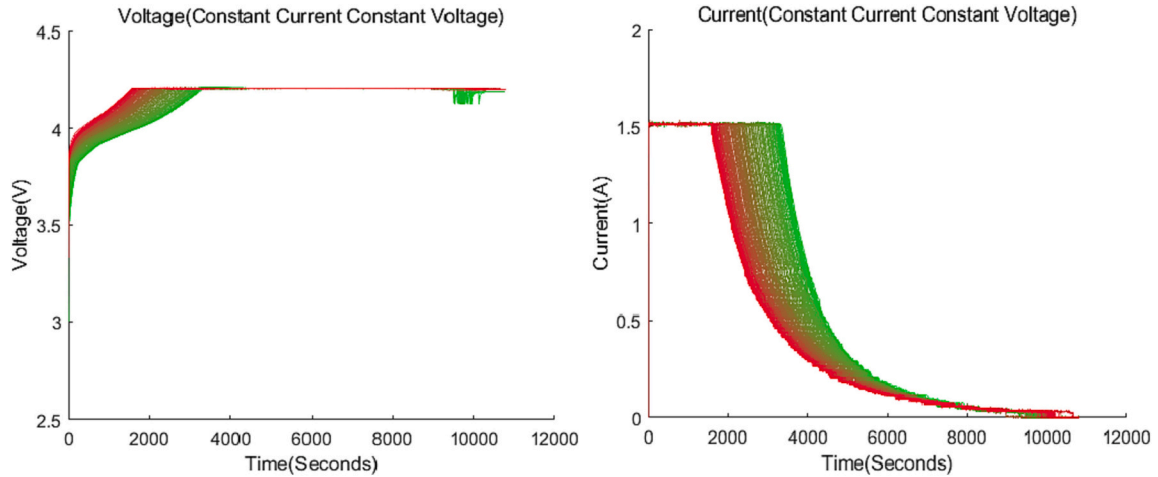


Fig. 4. Voltage curve and the current curve with the time of #5 battery.

is increased because electrode polarization is increased which would accelerate the speed of charge voltage arriving at the charge cutoff voltage.

From (9), the voltage could be obtained by battery capacity. In each cycle, the parameters could be identified based on data to fit the capacity and voltage without initial value. With direct-measured data, the noise such as measurement error and frequency sampling in voltage can be filtered by the OCV calibration model. Moreover, the battery model could describe the properties of the battery such as the phase transition

and staging phenomena with expert model-based knowledge to provide more reliable and higher interpretable features. Hence, the model-based health features could better reflect the state of the battery.

Battery models could be built to describe battery characteristics. However, accurate SOH estimation with models under the change of the operation condition is still a challenge because the battery models cannot be well adapted to all conditions. Data-driven methods could exploit the data to adapt to new conditions. Thus, designing a data-driven method based on battery models to improve estimation

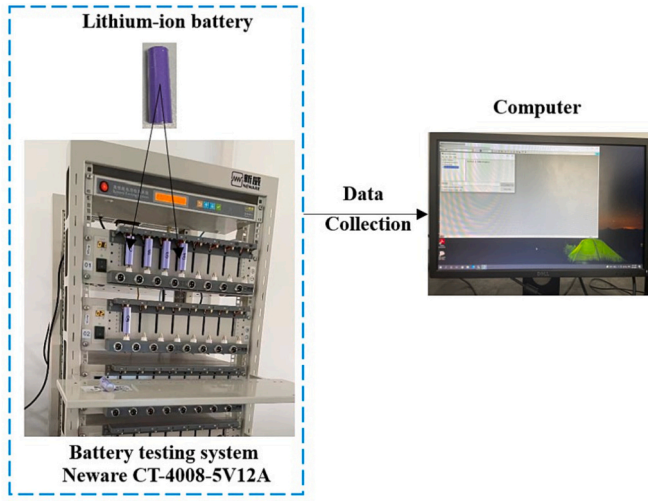


Fig. 5. Battery testing environment.

Table 1
Detailed description of batteries.

Condition	Battery No.	Temperature	Charging current	End voltage	Discharging current
1	#5	24 °C	1.5A	2.7 V	2A
2	#6	24 °C	1.5A	2.5 V	2A
3	#7	24 °C	1.5A	2.2 V	2A
4	#29	43 °C	1.5A	2.0 V	4A
5	#30	43 °C	1.5A	2.2 V	4A
6	#31	43 °C	1.5A	2.5 V	4A
7	FST	23 ± 2 °C	0.75A	2.5 V	1.5A

performance under different operation conditions with fewer data is necessary for the battery system.

3. Model-data fusion domain adaptation

3.1. Introduction of transfer learning

Transfer learning is a data-driven method to adapt the knowledge from the source domain to the target domain. The battery is always

working under different operation conditions and accurate labeled SOH requires special instruments which cannot always be obtained. As a result, labeled SOH data would be insufficient under various operation conditions. Hence, designing a transfer learning method to exploit sufficient data from different operation conditions as source domain is desirable for SOH estimation. Through reducing the probability distribution discrepancy between the dataset of the source domain and the target domain, the knowledge from the source domain could be transferred to the target domain in SOH estimation with a small amount of data. Transfer learning has been widely applied in solving NLP, image processing and speech process problems [35]. In recent years, transfer learning has been applied to improve the battery SOH estimation under

Table 2
The estimation errors of SOH under different methods.

Cell	Method	MAE	MARE(%)	RMSE
#6	Proposed method	0.0063	0.93	0.0088
	GPR	0.0174	2.61	0.0237
	Neural Network	0.0208	2.97	0.0256
	SVR	0.0240	3.54	0.0270
	Fine-tuning	0.0116	1.78	0.0159
	Domain adaptation	0.0115	1.58	0.0135
#7	Proposed method	0.0053	0.67	0.0071
	GPR	0.0334	4.47	0.0428
	Neural Network	0.0257	3.46	0.0342
	SVR	0.0263	3.54	0.0340
	Fine-tuning	0.0190	2.52	0.0246
	Domain adaptation	0.0141	1.81	0.0152
#29	Proposed method	0.0031	0.37	0.0038
	GPR	0.0133	1.48	0.0125
	Neural Network	0.0076	0.90	0.0086
	SVR	0.0061	0.72	0.0069
	Fine-tuning	0.0061	0.73	0.0069
	Domain adaptation	0.0037	0.44	0.0050
#30	Proposed method	0.0049	0.61	0.0058
	GPR	0.0115	1.42	0.0123
	Neural Network	0.0093	1.15	0.0109
	SVR	0.0082	1.02	0.0108
	Fine-tuning	0.0088	1.08	0.0099
	Domain adaptation	0.0078	0.94	0.0088
#31	Proposed method	0.0060	0.69	0.0086
	GPR	0.0110	1.27	0.0128
	Neural Network	0.0114	1.32	0.0131
	SVR	0.0097	1.14	0.0120
	Fine-tuning	0.0086	1.00	0.0106
	Domain adaptation	0.0073	0.84	0.0115

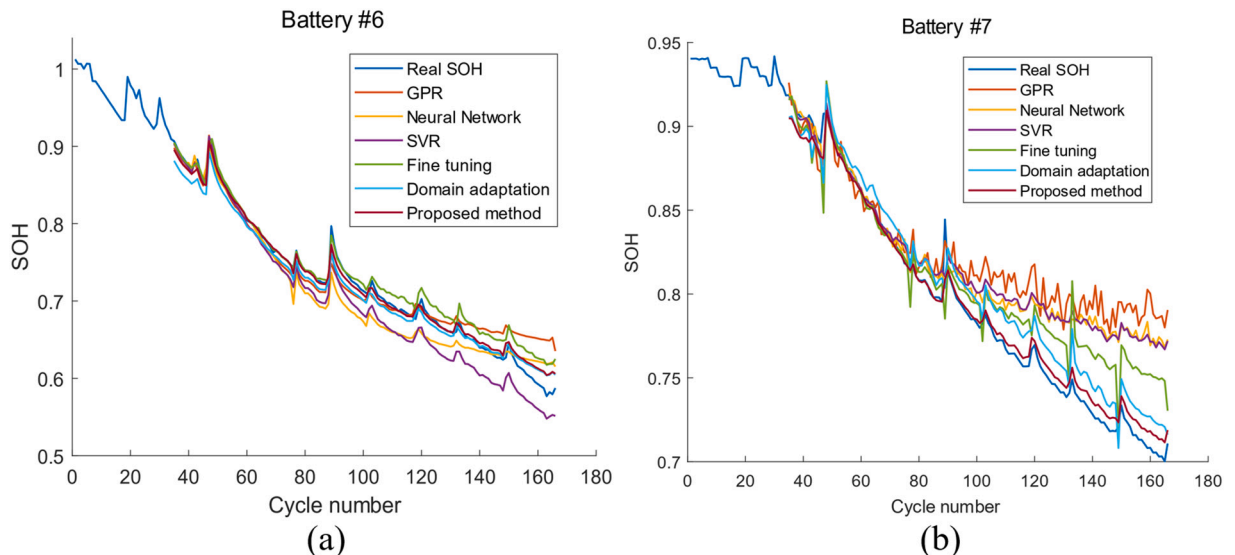


Fig. 6. (a) The comparison of SOH estimation results for battery #6. (b) The comparison of SOH estimation results for battery #7.

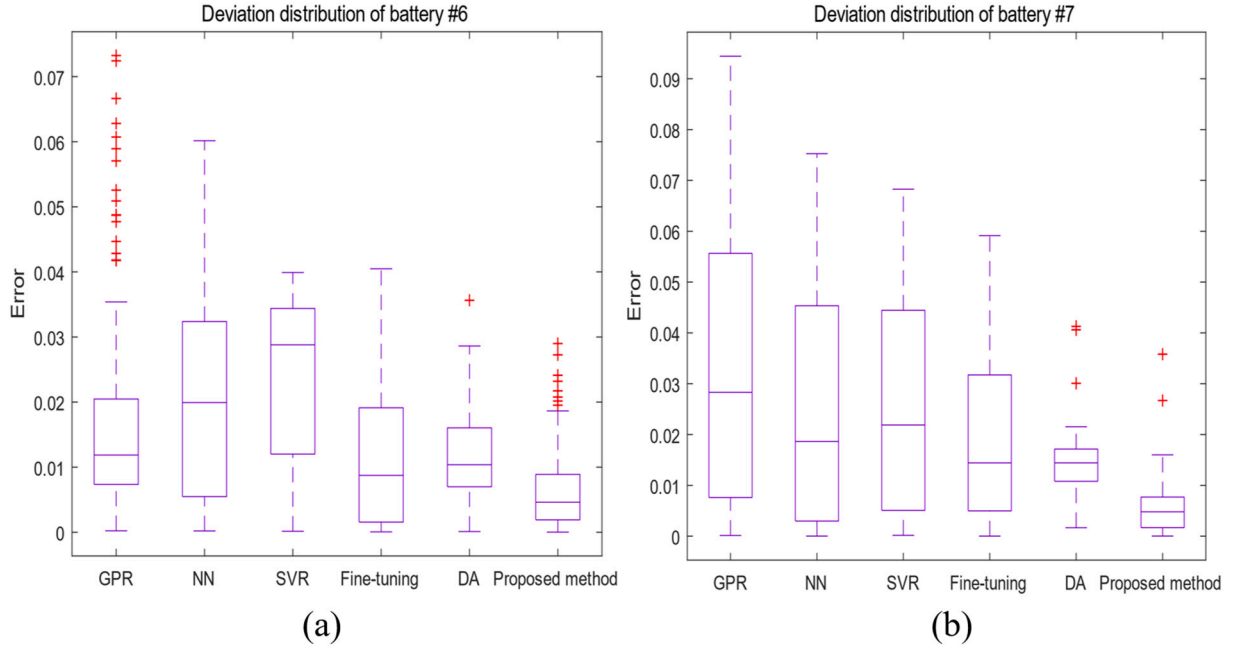


Fig. 7. (a) Deviation distribution of results by different estimation methods for battery #6. (b) Deviation distribution of results by different estimation methods for battery #7.

different operation conditions. Domain adaptation is a representative transfer learning method for improving estimation with a small amount of data under varying operation conditions by solving the distribution discrepancy problem. In order to evaluate the discrepancy between the source domain and target domain, some domain distance measures are proposed and MDD is a representative to measure the difference of distributions. With adversarial learning and rigorous generalization bounds, MDD could tailor to the distribution comparison with the asymmetric margin loss. Hence, the difference in distributions could be well measured which contributes to transferring more useful knowledge from the source domain to the target domain with fewer data. Due to the superiority discrepancy measurement, the MDD-based domain adaptation method is desirable for few-shot learning. Considering the collected data on battery features and battery cycles is not large, the MDD-based domain adaptation method is considered in this paper and the basic framework is shown in Fig. 2.

From Fig. 2, the minimization problem based on the margin loss and MDD could be expressed as:

$$\min_{f \in F} L_{all} = \text{err}_{D_s}^{(\rho)}(f) + d_{f,F}^{(\rho)}(\widehat{D}_s, \widehat{D}_t) \quad (10)$$

where $\widehat{D}_s$ denotes the source sample, $\widehat{D}_t$ denotes the target sample, $\text{err}_{D_s}^{(\rho)}(f)$ denotes the source domain margin error, $d_{f,F}^{(\rho)}(\widehat{D}_s, \widehat{D}_t)$ denotes the empirical MDD, f denotes the scoring function, F denotes the hypothesis space.

The margin error is defined by:

$$\text{err}_D^{(\rho)}(f) = E_{x \sim \widehat{D}} \Phi_\rho \circ \rho_f(x, y) = \frac{1}{n} \sum_{i=1}^n \Phi_\rho(\rho_f(x_i, y_i)) \quad (11)$$

where $\circ$ denotes the function composition and Φ_ρ denotes the defined linear piecewise function, x denotes input sample and $x \in R^{n \times g}$, y denotes output sample and $y \in R^n$, x_i denotes i -th features from input sample and $x_i \in R^{1 \times g}$, y_i denotes i -th output, n denote number of samples, g denotes the dimension of features.

The empirical MDD is expressed as below:

$$d_{f,F}^{(\rho)}(\widehat{D}_s, \widehat{D}_t) = E_{x^t \sim \widehat{D}_t} L(f(\psi(x^t)), f(\psi(x^t))) - \gamma E_{x^s \sim \widehat{D}_s} L(f(\psi(x^s)), f(\psi(x^s))) \quad (12)$$

where x^s denotes the input sample from source domain and $x^s \in R^{ns \times g}$, ns denotes number of samples from source domain, x^t denotes the input sample from target domain and $x^t \in R^{nt \times g}$, nt denotes number of samples from target domain, L and L' denote loss functions, ψ denotes the feature extract representation, f denotes the auxiliary function, f denotes the mapping function, γ denotes margin parameter.

Based on the framework in Fig. 2, MDD is designed for distribution comparison with asymmetric margin loss. Thereby, the probability distribution discrepancy between the dataset of the source domain and the target domain could be reduced. Thus, the insufficient data in the target domain could be trained to obtain the desired results by transferring the knowledge from sufficient data of the source domain.

Nevertheless, the unfavorable noise from raw data, the requirement for reliable and interpretable battery features, and the designing of the layer-reduced feature extractor for the commercial battery are still problems for the further improvement of domain adaptation under the different operation conditions with fewer data. In order to improve the estimation performance, a model-data fusion domain adaptation is designed to estimate the battery SOH in this paper.

3.2. Model-data fusion domain adaptation

For the battery system, mechanism models and equivalent models are built based on model-based knowledge. With the battery models, noise-reduced, reliable and informative features could be generated based on the direct-measured raw data. Besides, the generated features are more interpretable which is beneficial for designing the layer-reduced feature extractor of domain adaptation. Therefore, the problems of unfavorable noise, the requirement for reliable and interpretable features, and the designing of layer-reduced feature extractor could be solved by battery models. Thus, the battery models can be fused to domain adaptation and the framework of model-data fusion domain adaptation is shown in Fig. 3.

From Fig. 3, the battery models could be regarded as part of the

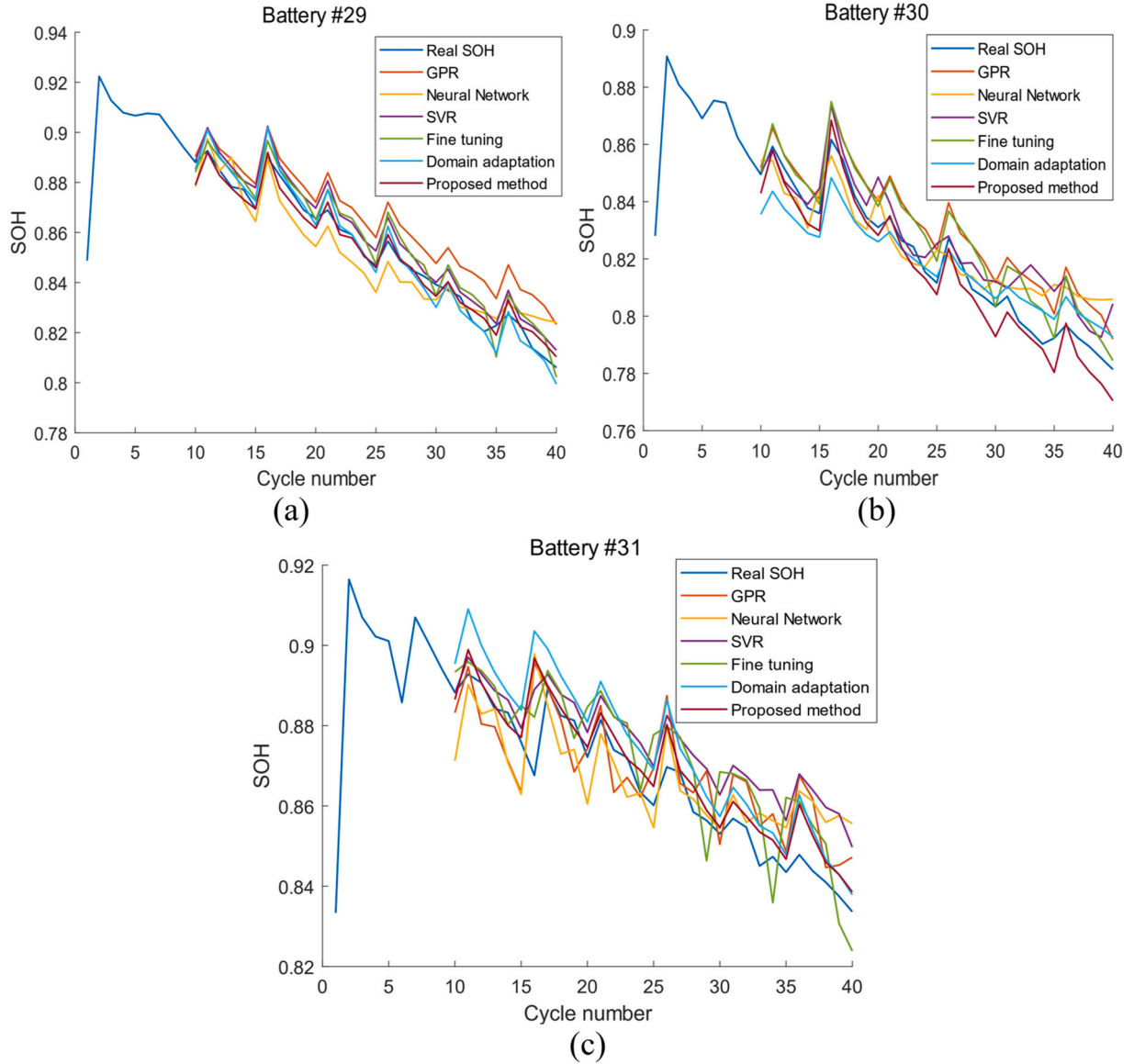


Fig. 8. (a) The comparison of SOH estimation results for battery #29. (b) The comparison of SOH estimation results for battery #30. (c) The comparison of SOH estimation results for battery #31.

common feature extractor. Different battery models are built to provide health features. With the reliable and interpretable features extracted by the battery models, the simplified feature extractor based on the convolutional neural network (CNN) framework could be introduced to further extract the features. For our model-data fusion domain adaptation, the feature extractor is consisted of the convolutional layer, batch layer, pool layer and flatten layer. The rectified linear unit layer is not considered owing to the desirable features of the constructed battery models. In the convolutional layer, the process could be illustrated as follows:

$$C_i = \text{CONV} \left(\sum H \times k_i + b_i \right) \quad (13)$$

where H denotes a given health features and $H \in R^{1 \times g}$, g denotes the dimension of features, k_i denotes i -th convolutional kernel, b_i denotes i -th bias, CONV denotes the active function.

After the convolutional operation, the batch layer is introduced for normalization and the pool layer is implemented to reduce the dimension. Besides, the flatten layer is implemented to reduce dimension size. The overall feature extract representation is defined by ψ .

With the extracted features, the MDD is introduced in this paper to reduce the probability distribution discrepancy between the source domain and target domain. The overall loss could be defined as:

$$L_{all} = L_s + \beta L_{da} \quad (14)$$

where L_s denotes the source loss, L_{da} denotes MDD based domain adaptation loss.

In this paper, the mean square error (MSE) is used to evaluate the source loss which is shown as:

$$L_s = \frac{1}{N_{source}} \sum_{i=1}^{N_{source}} (SOH_{source}^i - SOH_{estimated}^i)^2 \quad (15)$$

where H_i^{source} denotes the i -th health features in the source domain, SOH_{source}^i denotes i -th true SOH in the source domain, $SOH_{estimated}^i$ denotes i -th estimated SOH in the source domain, N_{source} denotes the training sample size in the source domain.

Loss L_{da} could be obtained by:

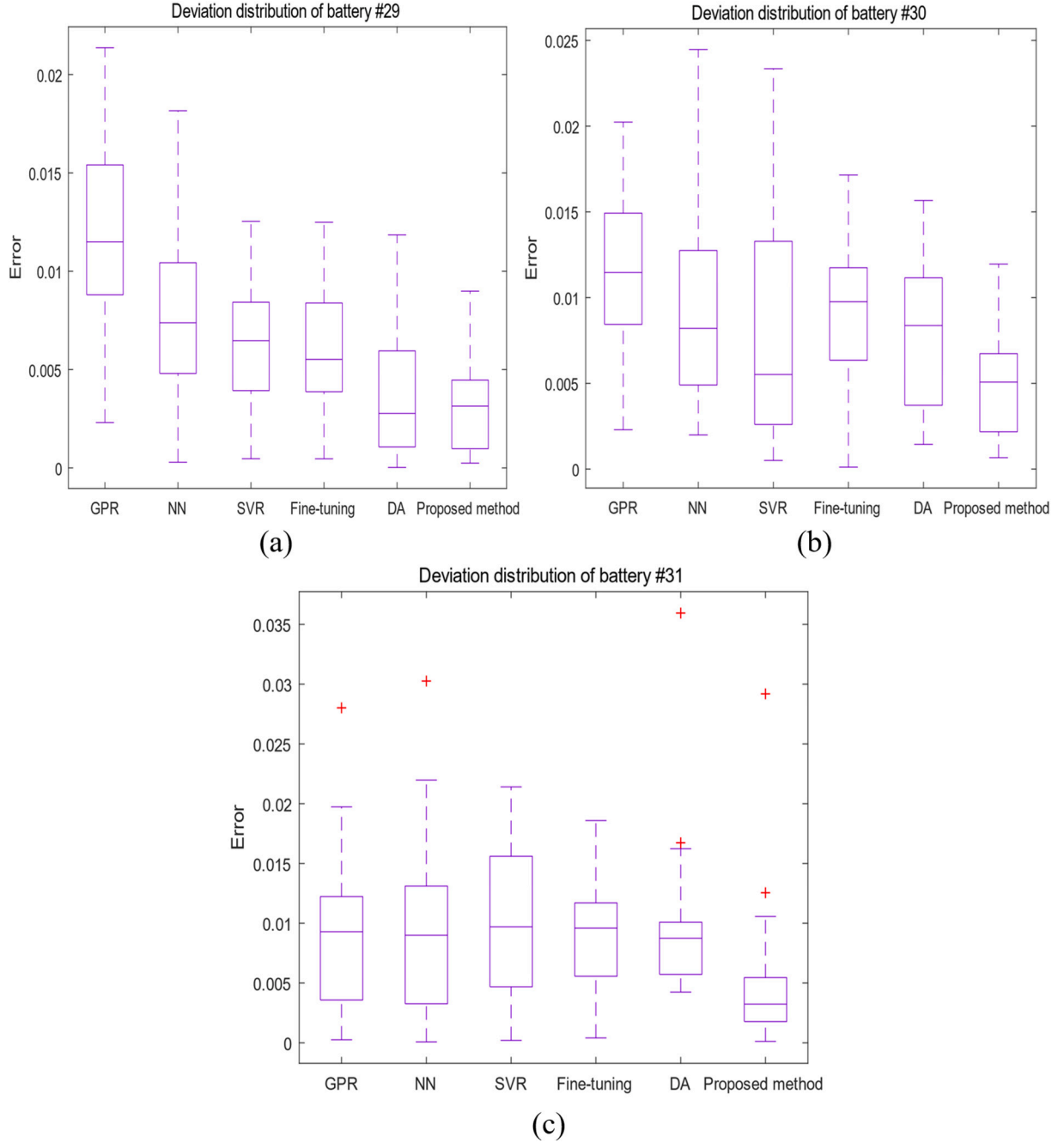


Fig. 9. (a) Deviation distribution of results by different estimation methods for battery #29. (b) Deviation distribution of results by different estimation methods for battery #30. (c) Deviation distribution of results by different estimation methods for battery #31.

$$L_{da} = \frac{1}{N_{\text{target}}} \sum_{i=1}^{N_{\text{target}}} |f'(\psi(H_i^{\text{target}})) - f(\psi(H_i^{\text{target}}))| - \frac{\gamma}{N_{\text{source}}} \sum_{i=1}^{N_{\text{source}}} |f'(\psi(H_i^{\text{source}})) - f(\psi(H_i^{\text{source}}))| \quad (16)$$

where f denotes scoring function, f' denotes auxiliary scoring function, N_{source} and N_{target} denotes the training sample size in the source domain and target domain, H_i^{source} and H_i^{target} denotes i -th health features in the source domain and target domain.

With L_{da} and L_s , the difference in marginal distribution could be reduced. Besides, different batteries operated under different operation conditions may result in a difference in conditional distribution. To improve the estimation performance, a few data in the target domain

could be introduced if a few data are available from examination and battery tests to reduce the difference in conditional distribution. Hence, loss in the target domain L_t is given by:

$$L_t = \frac{1}{N_{\text{target}}} \sum_{i=1}^{N_{\text{target}}} (SOH_{\text{true}}^i - SOH_{\text{estimated}}^i)^2 \quad (17)$$

where SOH_{true}^i denotes the i -th true SOH and $SOH_{\text{estimated}}^i$ denotes the i -th estimated SOH in target domain, N_{source} denotes the training sample size in the target domain.

The final overall loss could be given by:

$$L_{\text{all}} = L_s + \alpha L_t + \beta L_{da} \quad (18)$$

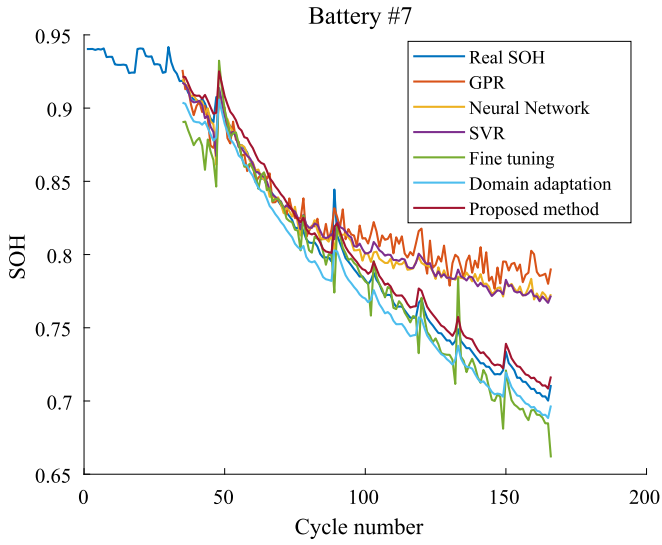


Fig. 10. The comparison of SOH estimation results.

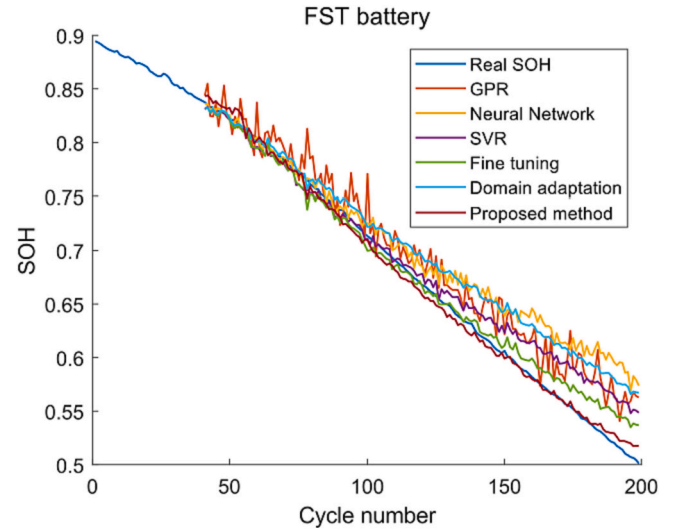


Fig. 12. The comparison of SOH estimation results.

Table 3

The estimation errors of SOH under different methods.

Cell	Method	MAE	MARE(%)	RMSE
#7	Proposed method	0.0075	0.94	0.0080
	GPR	0.0334	4.47	0.0428
	Neural Network	0.0257	3.46	0.0342
	SVR	0.0263	3.54	0.0340
	Fine-tuning	0.0125	1.58	0.0178
	Domain adaptation	0.0124	1.57	0.0129

Table 4

The estimation errors of SOH under different methods.

Cell	Method	MAE	MARE(%)	RMSE
FST	Proposed method	0.0059	0.90	0.0070
	GPR	0.0247	4.05	0.0298
	Neural Network	0.0292	5.06	0.0394
	SVR	0.0168	2.89	0.0227
	Fine-tuning	0.0103	1.72	0.0135
	Domain Adaptation	0.0276	4.69	0.0344

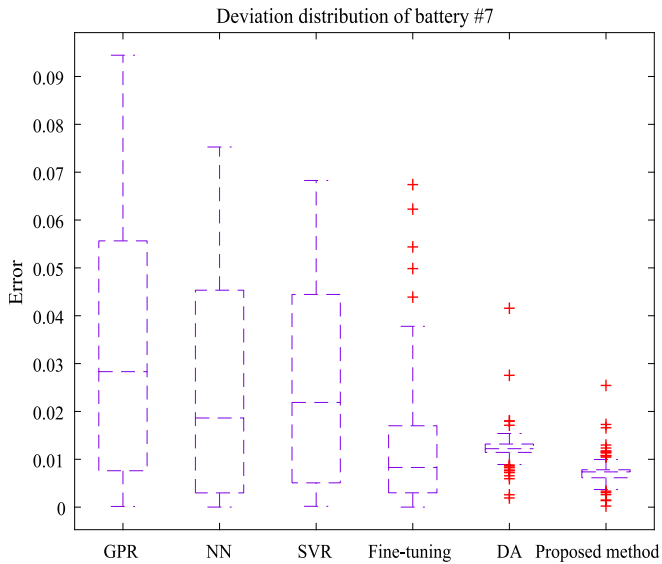


Fig. 11. Deviation distribution of results by different SOH estimation methods.

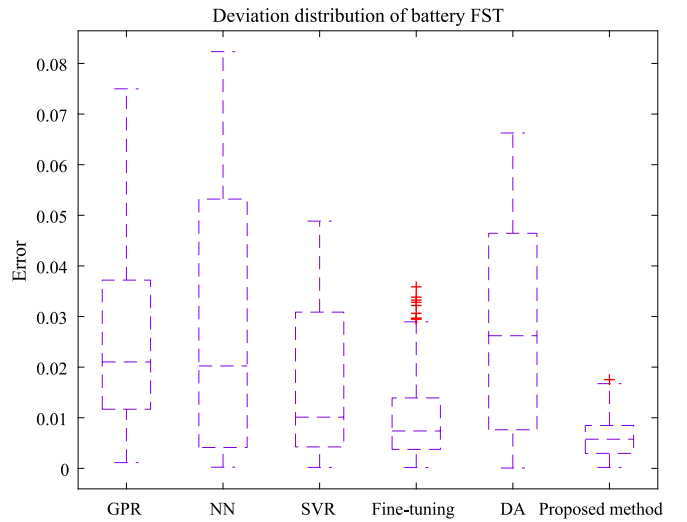


Fig. 13. Deviation distribution of results by different SOH estimation methods.

where α and β are both the non-negative trade-off parameter to balance the different loss terms, α is 0 when SOH label in target domain is unavailable, α and β are selected as 1 in this paper.

Through introducing the domain adaptation loss, the output based on the target domain could exploit the knowledge from the source domain. Therefore, the mapping of the target domain is trained with fewer data through learning from the source domain. Besides, model-based features could contribute to transferring knowledge. Hence, the SOH estimation performance in the target domain could be improved

with the framework from Fig. 3. The steps of the proposed method are presented as algorithm 1:

Algorithm 1. Model-data fusion domain adaptation.

Input: the source dataset $V_{so}, I_{so}, t_{so}, SOH_{so}$ and target dataset $V_{ta}, I_{ta}, t_{ta}, SOH_{ta}^r$.
 Build the battery model to generate model based health features.
 Initialize network parameters.
 While less than training epochs **do**:
 Based on the generated health features from model to calculate the L_s and L_t ;
 Compute the domain adaptation loss based on MDD L_{da} ;
 Compute the total loss $L_{all} = L_s + \alpha L_t + \beta L_{da}$;

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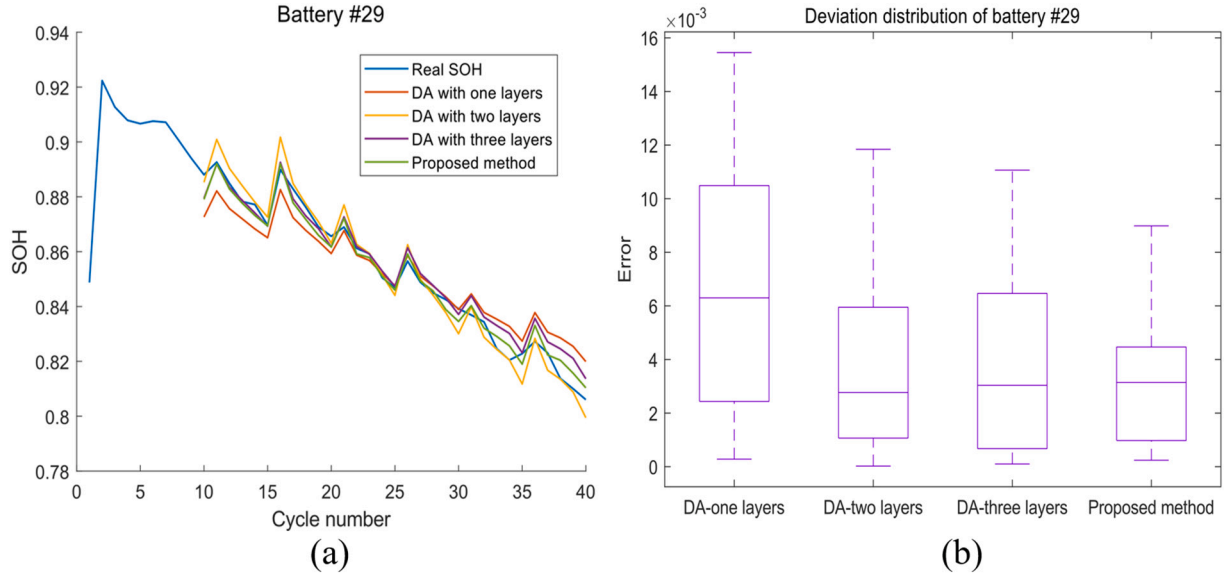


Fig. 14. (a) The comparison of SOH estimation results. (b) Deviation distribution of the results.

Table 5

The estimation errors of SOH under different methods.

Cell	Method	MAE	MARE(%)	RMSE
Battery #29	DA with one layers	0.0068	0.80	0.0082
	DA with two layers	0.0037	0.44	0.0050
	DA with three layers	0.0038	0.45	0.0051
	Proposed method	0.0031	0.37	0.0038

(continued)

Optimize the parameters based on gradient reversal to minimize the total loss.
Output: The model-data fusion domain adaptation.

3.3. Analysis of superiority of model-data fusion domain adaptation

In this part, the superiority of model-data fusion domain adaptation is analyzed.

Firstly, the noise could be filtered by the battery model. For the lithium-ion battery, noise is inevitable from the measurement sensor and frequency sampling. However, the accuracy of estimation is sensitive to noise for fewer training data. Since battery models such as the OCV

Table 6

The estimation errors of SOH with IC feature and without IC feature.

Battery	IC feature	MAE	MARE(%)	RMSE
#7	Without	0.0060	0.77	0.0078
	With	0.0053	0.67	0.0071
FST	Without	0.0059	0.90	0.0070
	With	0.0078	1.17	0.0089

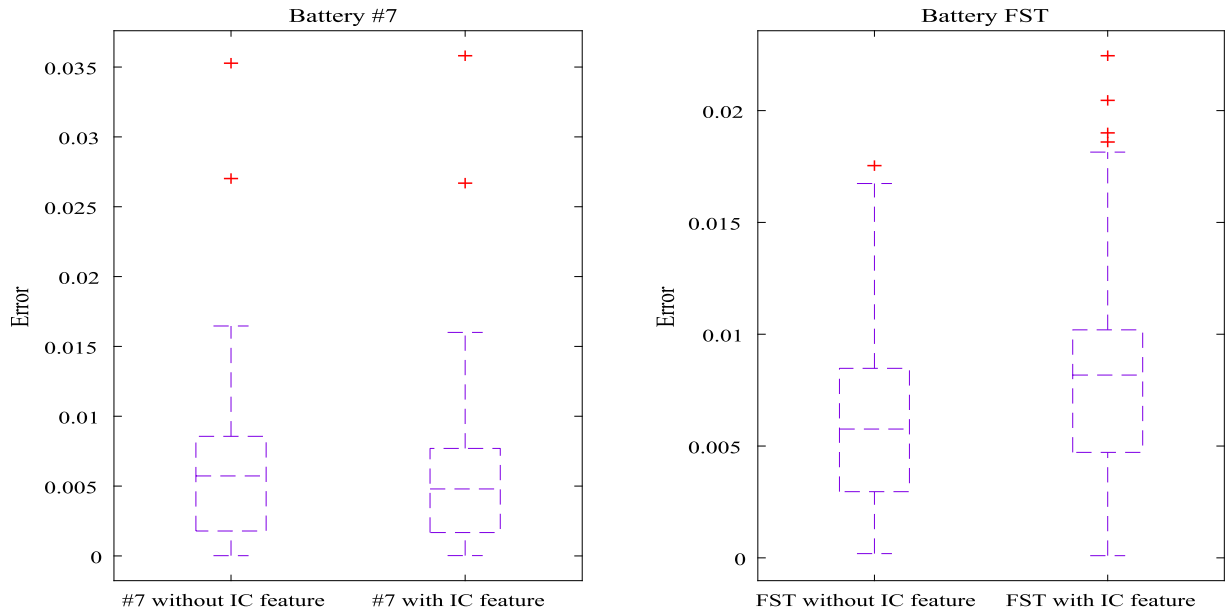


Fig. 15. The comparison of the proposed method with the IC feature and without the IC feature.

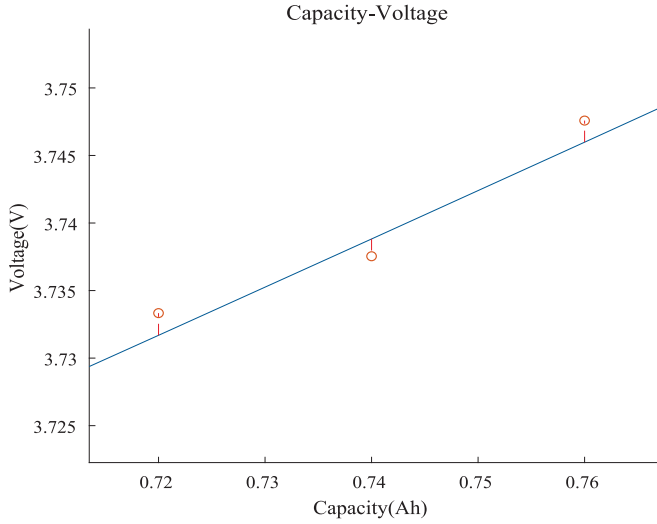


Fig. 16. Illustration of noise and curve filtering from the model of the FST battery.

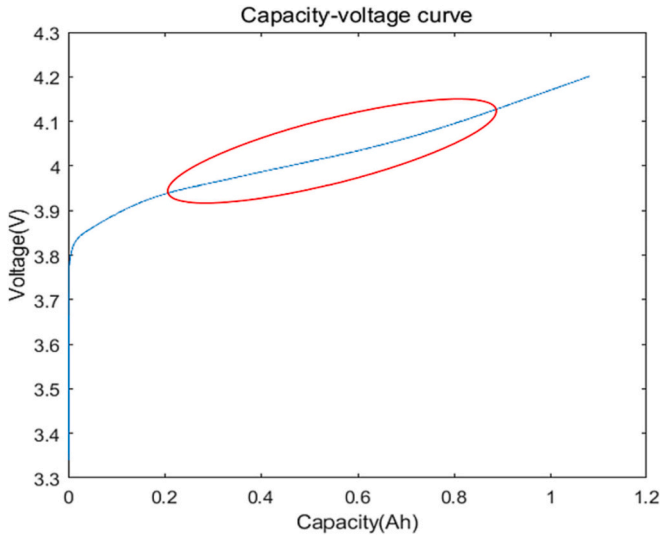


Fig. 17. Capacity-voltage curve during the period of constant current.

calibration model could better reflect battery characteristics such as lithiation and delithiation of the active material and phase transformation, the capacity-voltage curve could be much smoother and could be better approximated with the actual capacity-voltage curve. Thus, the voltage could be derived from the model to reduce the noise. With noise-reduced and reliable health features such as voltage, accuracy and generalization of model-data fusion domain adaptation could be improved even with fewer data.

Secondly, the features could be provided by the battery model which is a type of feature engineering. Various features based on the models such as the work model, incremental capacity model, and OCV calibration model could be generated from the features of voltage, current, time and temperature. Through model-based knowledge and prior mathematical analysis, generated features could be interpreted by the battery characteristics such as thermodynamic, electromotive force, lithiation and delithiation. Therefore, the features from various battery models in the source domain and the target domain are more reliable and interpretable which contributes to transferring more useful information from the source domain to the target domain. Hence, the model-based generated features are sufficient for providing more interpretable

and reliable information to train our method to explore SOH degradation under different operation conditions.

Thirdly, features with direct and strong relationships to SOH could be provided by models to simplify the layer of the feature extractor. Considering the model-based features are generated by the mathematical derivation and prior experience of numerous experiments, the features would be more relevant and direct to the SOH. For example, the battery work could be measured by the transformation of battery energy. During the charging process, the electrical energy would convert to chemical energy. From experiments of [36], the work would be increased at the same time period during the charging process with the SOH degradation. The reason is the reduction of active materials area and isolation of electrode areas which result in increasing internal resistance. Therefore, the voltage would arrive faster at the rated voltage as the degradation. With the model-based features, generated features are explained and are strongly correlated with SOH degradation. Hence, the complex common feature extractor is not required to extract the information from direct-measured raw features. Thus, model-data fusion domain adaptation could exploit model-based features to improve the performance with simplified networks.

4. Experiment

4.1. Introduction of battery dataset

To evaluate the effectiveness of the proposed model-data fusion domain adaptation, the battery of NASA Prognostics Center of Excellence (PCoE) and our experiment on the FST 18650 lithium-ion battery are applied [37]. The charging strategy is constant current-constant voltage. The voltage curve and the current curve with the time of the #5 battery in the whole life cycle are shown in Fig. 4.

For the NASA dataset, batteries #5–7 are operated at 24 °C. During the constant current mode, the batteries are charged at 1.5A until the voltage reaches 4.2 V. During the constant voltage mode, the batteries are kept at the constant voltage until the charging current reaches 20 mA. Under the discharging cycle, three batteries are discharged at 2A until the voltage drop to 2.7 V, 2.5 V and 2.2 V. For batteries #29–31, the batteries are operated at 43 °C. Under the charging cycle, the charging strategy is the same as #5–7. Under the discharging cycle, the batteries are discharged at 4A until the voltage dropped to 2.0 V, 2.2 V and 2.5 V. The rated capacity of the NASA battery is 2 Ah.

FST battery is a NCM full battery and the experiment is carried out at the Neware battery testing system and the testing environment is shown in Fig. 5.

In our experiment, the battery is charged at 0.75A until the voltage reaches 4.2 V under the constant current mode. Under the constant voltage mode, the battery is kept at the constant voltage until the charging current reaches 50 mA. During the discharging cycle, the discharging current is kept at 1.5A until the voltage is dropped to 2.5 V. The rated capacity of the FST battery is 2.5 Ah.

A detailed description of the NASA batteries and FST battery is shown in Table 1.

4.2. Evaluation criteria

To evaluate the accuracy of SOH estimation, the mean absolute error (MAE), the root-mean-square error (RMSE) and the mean absolute relative error (MARE) are used as criteria which are defined as:

$$MAE = \frac{1}{N} \sum_{i=1}^N |SOH_i^{real} - SOH_i^*| \quad (19)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (SOH_i^{real} - SOH_i^*)^2}{N}} \quad (20)$$

$$MARE = \frac{1}{N} \sum_{i=1}^N \frac{|SOH_i^{real} - SOH_i^*|}{SOH_i^{real}} \times 100\% \quad (21)$$

where SOH_i^{real} is the i -th real battery SOH from the experiment, SOH_i^* is i -th estimated SOH, N is the amount of measured data.

4.3. Case study 1: NASA datasets

For the NASA dataset, two conditions are considered where the end voltage is different for batteries #5 and #6-7; and the temperature and the end voltage are different for batteries #5 and #29-31. For the pairs of batteries #5 and #6-7, the source domain is battery #5 and the target domain is battery #6 or #7. Besides, the features are provided by the work model, the OCV calibration model with the order of 3 and the incremental capacity model. The work model provides a feature of the integrated power with respect to time, the OCV calibration model provides 5 features of the voltage at the 5 same capacity, and the incremental capacity model provides 2 features of mean and standard deviation from the Gaussian distribution. For the pairs of batteries #5 and #29-31, the source domain is battery #5 and the target domain is battery #29, #30 or #31. Besides, the features are provided by the work model and OCV calibration model with the order of 3. Considering the characteristic of the incremental capacity model is different between batteries #5 and #29-31, the incremental capacity model is not applied. The work model provides a feature of the integrated power with respect to time, the OCV calibration model provides 7 features of the voltage at the 7 same capacity. For each cell, the first 20% of the health feature data and 10% of real SOH data in the target domain are used for training model-data fusion domain adaptation. To demonstrate the effectiveness of the proposed method, BP Neural Network, support vector regression (SVR), Gaussian process regression (GPR), convolutional neural network (CNN) with fine-tuning, and domain adaptation without battery model are presented for comparison. Considering the irregular degradation of batteries #6-7, the first 40% of the target domain data are implemented for training BP Neural Network, SVR and GPR. For batteries #29-31, the first 20% of the target domain data are implemented for training BP Neural Network, SVR and GPR. For both batteries #6-7 and #29-31, the first 20% of the target domain data are applied for training the convolutional neural network with fine-tuning. The first 20% of the health feature data and 10% of real SOH data are applied for domain adaptation without the battery model for both batteries #6-7 and #29-31. The estimation results for batteries #6-7 are shown in Fig. 6 and Table 2. The distributions of the estimation results for the SOH by different methods are shown in Fig. 7. The estimation results for batteries #29-31 are shown in Fig. 8 and Table 2. The distributions of the estimation results for the SOH by different methods are shown in Fig. 9.

Besides, the FST battery is also applied as the source domain for comparison. The source domain is battery FST and the target domain is battery #7. The estimation results for #7 are shown in Fig. 10 and Table 3. The distributions of the estimation results for the SOH by different methods are shown in Fig. 11.

4.4. Case study 2: FST datasets

For our FST dataset, the source domain is the #5 battery and the target domain is the FST battery from our experiment. For the two batteries, the temperature, the type of battery and the end voltage are different. The work model provides a feature of integrated power with respect to time, the OCV calibration model provides 7 features of the voltage at the 7 same capacity. For our FST battery, the first 20% of health feature data and 10% of real SOH data are used for training model-data fusion domain adaptation and domain adaptation without the battery model. To demonstrate the superiority of the proposed method, the BP Neural Network, SVR, GPR, and CNN with fine-tuning are presented to make a comparison. The first 20% of the data are

used for training. The estimation results are shown in Fig. 12 and Table 4. The distributions of the estimation results for the SOH under different methods are shown in Fig. 13.

5. Results analysis and discussion

5.1. Results analysis

From the results in the last section, the accuracy of the proposed method is among the best in all the comparison methods. With the data in the source domain, the knowledge of SOH degradation could be transferred to estimate SOH in the target domain with a small sample. Therefore, the domain adaptation and CNN with fine-tuning are better than GPR, BP Neural Network and SVR for batteries #6-7 which can be seen in Fig. 6, Fig. 7 and Table 2. In addition, the domain adaptation and CNN with fine-tuning are also better than GPR, BP Neural Network and SVR for battery #7 with FST battery as the source domain which can be seen in Fig. 10, Fig. 11 and Table 3. Due to the difference in battery type, the performance with battery FST as the source domain is worse than the performance with battery #5 as the source domain because the characteristics of battery degradation are changed. Besides, the domain adaptation without battery model and CNN with fine-tuning show no obvious better than GPR, BP Neural Network and SVR from Fig. 8, Fig. 9, Fig. 12, Fig. 13, Table 2 and Table 4 since the characteristics of battery degradation are changed for batteries #29-31 and FST battery. For batteries #29-31, the variation of voltage with capacity is irregular during the prior phase of constant current mode which threatens the generalization of CNN with fine-tuning. For battery FST, the measurement noise and irregular variation of voltage during the end of the constant current mode result in the undesirable performance of domain adaptation without battery models. Therefore, SOH degradation information cannot be well extracted with the limited data and insufficient common feature extractor which results in undesirable SOH estimation results. For our method, the battery model could filter the noise and generate new features. The features which are more correlated with battery degradation are extracted for training based on the model-based knowledge. Hence, the proposed method is better than the domain adaptation without the battery model and CNN with fine-tuning.

5.2. Discussion

In this part, the reduction of layers for the common feature extractor, the reliability and interpretability of model-based features, and the raw data noise filter are discussed.

The features from the battery model provide strong SOH degradation correlated features through the model-based knowledge. Therefore, the convolutional layer could be reduced since the raw data are filtered by the model, and the rectified linear unit layer could be reduced since the features are more correlated with battery SOH. In this paper, battery #29 is illustrated to evaluate the reduction of layers with the proposed method. The domain adaptation without the battery model with one, two and three convolutional layers and batch layers, and the proposed method with one convolutional layer and batch layer are applied for SOH estimation. The results are shown in Fig. 14 and Table 5.

From Fig. 14 and Table 5, the proposed method is designed with one convolutional layer and batch layer which is better than the domain adaptation with one convolutional layer and batch layer which demonstrates the battery model could improve the estimation results of the proposed method. Similar results could be obtained by domain adaptation with two layers, domain adaptation with three layers and the proposed method because the sample is too small and the domain adaptation with two layers is enough to explore the trend of battery degradation. Nevertheless, the framework of the network of the proposed method is simpler than the domain adaptation with two layers which demonstrates the proposed method with the compression network could obtain similar results.

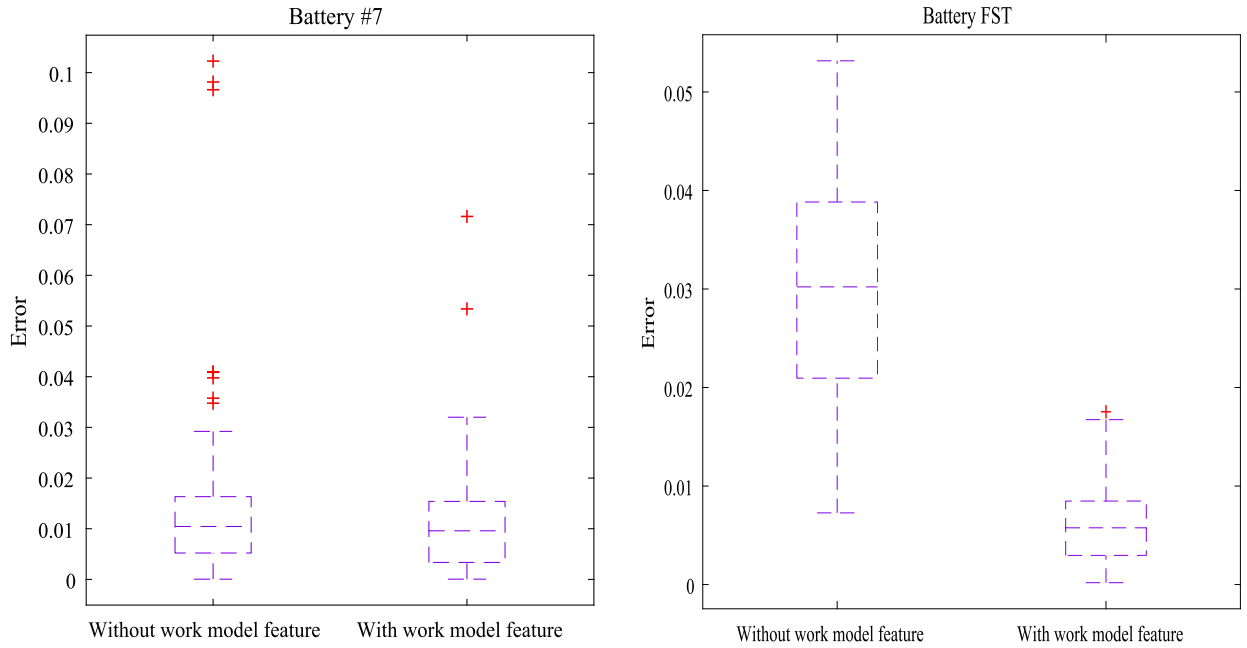


Fig. 18. The comparison of the proposed method with and without the feature from work model.

Besides, the noise-reduced, informative and interpretable features could be generated by the battery model. Therefore, the model-based different features are analyzed in this part.

Firstly, the incremental capacity (IC) could be obtained by capacity and voltage which could reflect the phase change of anode/cathode materials. The internal electrochemical reactions result in different peaks from the incremental capacity analysis. As the SOH degradation, the battery structure is changed. Hence, greater polarization is required for electrochemical reactions. As a result, the voltage is increased because of greater polarization needed to ensure the battery internal electrochemical reactions. Therefore, the position of the peak is shifted into higher voltage and the peak dQ/dV value is decreased which results in the increase of mean and variance value as battery health degradation. From the NASA dataset, batteries #5 and #6–7 are operated under similar working conditions where the characteristics of phase change are similar. Thus, the IC feature including mean and variance features is provided for training. However, for batteries #29–31 and our experiment on the FST battery. The change in temperature and the type of battery would result in different characteristics of phase change. Hence, the features based on incremental capacity without any treatment are not considered for training to avoid the degradation of the estimation performance. The comparisons are illustrated in Fig. 15.

From Fig. 15 and Table 6, the results of battery #7 with the IC feature are better than the results without the IC feature. The reason is the characteristics of phase change are similar between batteries #5 and #7. Hence, more features could provide more SOH degradation information for improving the estimation. Nevertheless, the results of FST battery with the IC feature are slightly worse than the results without the IC feature because the characteristics of phase change between the battery FST and battery #5 are different. The small sample in the target domain could contribute to the training of high performance proposed method.

Table 7

The estimation errors of SOH with and without the feature from work model.

Battery	Feature from work model	MAE	MARE(%)	RMSE
#7	Without	0.0069	0.86	0.0104
	With	0.0053	0.67	0.0071
FST	Without	0.0295	4.32	0.0314
	With	0.0059	0.90	0.0070

Nevertheless, the sample size is too small which would result in undesirable results since the SOH degradation information cannot be well extracted.

Secondly, the battery voltage is the crucial indicator for battery degradation and the accurate and reliable voltage feature is critical for estimating SOH with the small training sample size. With the battery model, the noise of voltage could be filtered which is analyzed below:

In our experiment, the measurement noise of our FST battery is $\pm 0.1\%$ and the noise should be filtered. The OCV calibration model could measure the phase transition and staging phenomena. Therefore, the noise filtering from the raw data is more reliable because of the battery knowledge. Hence, the curve is much smooth and more reliable which is shown in Fig. 16.

Besides, the battery polarization could be divided into three phases. From the expert knowledge, the variation of voltage can be seen in Fig. 17.

From Fig. 17, the variation of voltage in the ohmic polarization is more stable which could be seen in the red-marked area. Hence, the extracted voltage features are desirable for training with prior knowledge from modeling. For our FST battery, the electrolyte concentration of the electrode surface is reduced in the last phase of constant current mode resulting in concentration polarization. Hence, the concentration polarization is dominated which results in the more complex variation of voltage. Since the Arrhenius based OCV calibration model cannot well describe the phenomenon of concentration polarization, the Nernst equation would be considered in future work to generate more features. For the #29 battery with higher temperature, rising ionic conductivity and decreasing viscosity of electrolyte, as well as the aggravated side effects between the electrolyte and active materials, would result in the fluctuation of dynamic performance in the prior phase of the constant current mode. Hence, the variation of voltage is irregular because of the change in dynamic performance. The irregular voltage features are undesirable for measuring SOH degradation. To improve the estimation, the voltage features in the middle phase where ohmic polarization is dominated are more reliable for SOH estimation where the variation of voltage is more stable.

From the above analysis, the battery model could filter the noise and the model-based knowledge could be exploited to extract the feature which is more reliable and interpretable for SOH estimation under different operation conditions with a few data.

Finally, the work is calculated by integration with a period of time. During the battery degradation, the dynamic performance of anode and cathode materials would be undesirable which results in the reduction of active materials and isolation of electrode areas. Hence, the internal resistance is increased and the surface charge accumulation is accelerated. Therefore, the electrode polarization is increased which would accelerate the speed of charge voltage arriving at the cutoff voltage. Thus, the charge voltage is increased in the constant current mode where the work in the same period is increased as the SOH degradation. To illustrate the performance of extracted features from the work model, domain adaptation with and without the extracted feature from the work model are illustrated in Fig. 18 and Table 7.

From Fig. 18 and Table 7, the better estimation results could be obtained with the feature from the work model. The reason is the work model could reflect the variation of electrode polarization during the degradation which is more reliable and more interpretable for battery SOH estimation.

6. Conclusion

In this paper, a novel model-data fusion domain adaptation is proposed for estimating the SOH of the lithium-ion battery with fewer data under different operation conditions. The OCV calibration model, the working model and the modeling of incremental capacity are introduced to generate the features. Hence, the noise-reduced, reliable and interpretable features are applied for the feature extraction which could reduce the layer of the feature extractor. With the extracted model-based features, the MDD-based domain adaptation is implemented to reduce the probability distribution discrepancy. As a result, the SOH could be estimated with fewer data in the target domain through learning the knowledge from the source domain. The experimental results demonstrate that the proposed method is better than other methods because of model-based features and the transfer of knowledge from the source domain under different operation conditions. Besides, the results from the analysis also demonstrate the effectiveness of the proposed method with fewer layers of the feature extractor and interpretable features from the models.

Future work would further analyze mechanism models and empirical models to explore more reliable and interpretable battery features. Besides, the features would be extracted based on operation conditions to explore the SOH estimation with zero-shot learning.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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